

02 · Software Requirements Specification

Software Requirements Specification

Project	SDJ Editorial Portal — an editorial portal for the Science and Development Journal (SDJ), published by the College of Basic and Applied Sciences (CBAS), University of Ghana
Document	02 — Software Requirements Specification
Author	Roger Koranteng Obeng, student ID 22424140
Assessor	Prof. Solomon Mensah
Date	2026-08-12
Conformance	Adapted from IEEE 830-1998 and ISO/IEC/IEEE 29148:2018
Status	Authoritative. Where this document and the implementation disagree, the implementation governs and the disagreement is recorded (section 4.1, section 7).

Naming. The product specified here is the **SDJ Editorial Portal**. The repository (github.com/rogerkoranteng/ugjcs), hosting URLs, package names and infrastructure resource names retain the pilot's internal codename **UGJCS**; wherever `ugjcs` appears in a path, URL or identifier, it is that codename, not a separate system. The deployed system is a prototype built for an Advanced Software Engineering exam, not SDJ's official production system.

Revision note (2026-08-12, post-implementation). This document was originally authored before the API and frontend existed, when only the domain layer and the database persistence adapter were built. Section 2.1, section 2.4, section 4.3, section 5 and section 7 have since been revised against the finished, deployed system — the live API (<https://tsxsbf9rzp.us-east-1.awsapprunner.com>), confirmed against its own `/openapi.json`, the live frontend (<https://ugjcs-frontend.vercel.app>), the backend source under `backend/src/ugjcs/api/` and `frontend/src/app/`, the test suites under `backend/tests/`, and `docs/06-testing-report.md`. That first revision moved twelve requirements from planned to implemented-and-tested, refined six to "partially implemented" with a named gap, and left eleven genuinely not implemented.

A second revision on 2026-08-14 reconciled the document against two feature waves that landed after the original build. It moved eight further lines to implemented-and-tested, added six new requirements (FR-29 to FR-34) for capabilities the original set never anticipated, and corrected several rows that had begun to understate the system rather than overstate it. The current totals are in section 5. Requirement identifiers, MoSCoW priorities and the sections neither revision touched are unchanged from the original.

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of the SDJ Editorial Portal, a double-blind peer-review and editorial management platform built for the Science and Development Journal (SDJ), published by the College of Basic and Applied Sciences, University of Ghana. It is written for three audiences: the assessor judging requirements-engineering competence against this final project's rubric, a future maintainer who needs to know what the system is contracted to do, and the author's own implementation, which this document constrains. Requirements are stated so that each is individually testable; no requirement is expressed as an unmeasurable quality goal.

1.2 Scope

The portal manages SDJ's complete scholarly publishing lifecycle: account registration and role-based access; manuscript submission with file upload; automated post-upload processing; editorial screening; assisted reviewer assignment; structured double-blind review; editorial decisions and revision rounds; issue composition and publication; a public archive with search, citation export and metadata harvesting; and a tamper-evident editorial audit trail. Payment processing, copy-editing/typesetting, multi-journal tenancy, ORCID federation, real DOI registration, and plagiarism screening against external corpora are explicitly out of scope (design specification section 4.2). The full functional boundary is enumerated in section 5 of that specification and restated as testable requirements in section 3 of this document.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
SDJ	Science and Development Journal — the established CBAS journal this portal is built for
CBAS	College of Basic and Applied Sciences, University of Ghana — SDJ's publisher and the project's client
UGJCS	The pilot's internal codename for this system (see the naming note above) — survives only in package names, the repository name and hosting URLs
Double-blind	Neither author nor reviewer identity is disclosed to the other party during review
MoSCoW	Must/Should/Could/Won't — requirement prioritisation scheme
UCP	Use Case Points — the effort-estimation technique that produced the MoSCoW cut
FR / NFR	Functional requirement / non-functional requirement
UC	Use case, as enumerated in the effort-estimation document
RBAC	Role-based access control
COI	Conflict of interest
EiC	Editor-in-Chief
HoD	Head of Department — the design's shorthand for the college-leadership analytics consumer
BFF	Backend-for-Frontend — the Next.js server-side proxy pattern used for authenticated traffic
OAI-PMH	Open Archives Initiative Protocol for Metadata Harvesting
DOI	Digital Object Identifier (the portal issues DOI- <i>shaped</i> but unregistered identifiers — section 4.2)
TF-IDF	Term frequency-inverse document frequency, used for reviewer-manuscript matching
LSH	Locality-sensitive hashing, used for near-duplicate similarity screening
WCAG	Web Content Accessibility Guidelines
RFC 9457	Problem Details for HTTP APIs — the platform's error-response format
Hash chain	The append-only audit mechanism: each event hashes over its predecessor's hash and its own payload

1.4 References

- Design specification — `docs/superpowers/specs/2026-08-12-ugjcs-journal-platform-design.md`
- Effort estimation — `docs/03-effort-estimation.md`
- Technical debt register — `docs/04-technical-debt-register.md`
- Domain implementation — `backend/src/ugjcs/domain/{transitions,enums,policies,blinding,manuscript,hashchain,events}.py`
- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*
- ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*

1.5 Overview

Section 2 describes the product, its user classes and its operating constraints. Section 3 states each functional and non-functional requirement in testable form. Section 4 gives the manuscript lifecycle as implemented, the actor/use-case mapping and the authorisation matrix. Section 5 is the requirements traceability matrix, stated honestly for what is built versus planned. Section 6 explains the MoSCoW prioritisation and how the effort estimate drove it. Section 7 states the system's limitations plainly, cross-referenced to the technical debt register.

2. Overall description

2.1 Product perspective

The portal is a new, self-contained system built for the Science and Development Journal (SDJ), published by the College of Basic and Applied Sciences; it replaces the journal's current ad hoc process of email, shared drives and spreadsheets rather than integrating with an existing editorial system. It is a three-tier web application: a Next.js frontend (public archive, search, paper detail, and author/reviewer/editor/Editor-in-Chief screens, with all authenticated traffic proxied server-side as a Backend-For-Frontend), a FastAPI backend built as a hexagonal architecture with a framework-free domain core, and PostgreSQL and S3 as its persistence and document-storage infrastructure.

This is no longer a forward-looking description. The domain layer, the database persistence adapter, the application use-case layer, the FastAPI API layer (/auth , /admin , /billing , /manuscripts , /editorial , /editorial-certificate , /reviews , /archive , /people , /health , /ready) and the Next.js frontend are all built, tested and deployed: API at <https://tsxsb9rzp.us-east-1.awsapprunner.com> , frontend at <https://ugjcs-frontend.vercel.app> . Section 5's traceability matrix states, requirement by requirement, which of these are implemented and tested, which are partially built with a named gap, and which are not built at all.

Four things named in the original design remain unbuilt: OAI-PMH harvesting (FR-22), similarity screening (FR-20), notifications of any kind, and asynchronous post-upload processing. Redis and an ARQ worker were designed for that last one (design specification Section 7) but are absent from the delivered codebase, and nothing in the current system depends on them; FR-20 is the requirement that would have used them. Terraform-managed ECS/ALB/CloudFront was the original deployment design, and the actual deployment target differs (section 2.4, TD-14).

Six capabilities have no counterpart in the original requirement set, because they were identified after it was written. They are added as FR-29 to FR-34: article processing charges, the administrator console, decision certificates, anonymisation preflight, review deadlines, and self-service author registration.

2.2 Product functions — summary

- Register, authenticate and authorise users under five roles, several of which one account may hold concurrently.
- Accept manuscript submissions, validate and process them asynchronously, and produce an anonymised derivative for reviewers.
- Screen submissions and route them to review, revision, or desk rejection.
- Recommend reviewers by expertise match, exclude conflicts, balance workload, and let the editor accept, reject or override every recommendation.
- Collect structured, double-blind reviews and aggregate them into an editorial decision once a quorum is met.
- Support revision rounds between author and editor.
- Compose issues from accepted manuscripts and publish them to a public archive.

- Serve the archive to anonymous readers with full-text search, citation export and OAI-PMH harvesting.
- Record every editorial event in a tamper-evident, append-only log.
- Bill and settle an article processing charge once a paper is accepted.
- Administer accounts, roles, reviewer capacity and activation from a console.
- Let anyone verify a published paper's editorial history against a tamper-evident chain.

This list states intended product scope. It is not a claim that every bullet is built. Section 5's traceability matrix is the authoritative, requirement-by-requirement account of what is implemented today. Of the bullets above, asynchronous processing (section 2.1) and OAI-PMH harvesting (FR-22) are the two that are not built.

2.3 User classes and characteristics

User class	Description	Frequency of use	Technical proficiency required
Author	Submits and tracks manuscripts, responds to review	Occasional, bursty around submission and revision	Low — web form and file upload only
Reviewer	Evaluates blinded manuscripts against structured criteria	Occasional, invitation-driven	Low
Editor	Screens submissions, assigns reviewers, records decisions	Regular, throughout each submission's lifecycle	Moderate — must interpret matching output and audit trail
Editor-in-Chief	All Editor capability plus issue composition, publication and policy configuration	Regular; final authority	Moderate
Administrator	Manages accounts, roles and reviewer capacity	Infrequent	Low
Reader	Anonymous; discovers, reads, cites and downloads published work	Frequent, unauthenticated	Low
CBAS college leadership (secondary)	Consumes editorial throughput analytics	Infrequent	Low
Indexing service (external system)	Harvests metadata via a defined API	Automated, scheduled	N/A — machine actor

A single account may hold multiple roles simultaneously (design specification section 3); this is a deliberate design choice with a direct security consequence recorded in section 4.3 and Section 7 (an Author–Reviewer dual-role holder is not currently prevented from reviewing their own manuscript).

2.4 Operating environment

As deployed (superseding the design specification's ECS/ALB/CloudFront topology — see TD-14 for why, and for why no functional capability was lost by the substitution): Backend: Python 3.13, FastAPI, PostgreSQL 16, deployed as a container on AWS App Runner, reachable at <https://tsxsbf9rzp.us-east-1.awsapprunner.com> with a trusted TLS certificate on App Runner's own `*.awsapprunner.com` hostname (no registered domain is available, which is what the CloudFront design was originally solving; App Runner supplies the same guarantee directly). Frontend: Next.js 15 / TypeScript / Tailwind, deployed on Vercel at <https://ugjcs-frontend.vercel.app>. Storage: private S3 buckets, documents reachable only via short-lived pre-signed URLs. Client environment: any evergreen browser at 360 px width or above; readers require no account. Redis and an ARQ worker, specified for asynchronous post-upload processing, are not present in the delivered infrastructure — nothing currently deployed depends on them (section 2.1).

2.5 Design and implementation constraints

- A 48-hour solo development window (effort estimation section 8), which forced the MoSCoW cut in section 6 of this document.

- No registered domain — the CloudFront TLS strategy is not optional but architecturally necessary (design specification section 7.3).
- AWS and Vercel free/low-cost tiers; operating cost is targeted at USD 35–55/month.
- Domain layer must import no framework code, enforced by an import-linter contract in CI (NFR-13). **As delivered**, two contracts are enforced (`backend/.importlinter`): `domain-purity` (the constraint above) and `layers` (`api` → `infrastructure` → `application` → `domain` , dependencies point inward only) — both gate `make check` and therefore CI on every push and pull request.
- The domain layer is framework-free by construction; this is a testable constraint, not an aspiration.
- Synthetic seed data only — no real SDJ submissions or reviewer data are used.

2.6 Assumptions and dependencies

Assumptions (stated as assumptions — the project had no access to real SDJ operational data): a single journal instance, SDJ itself, suffices; reviewers participate without incentive mechanisms; SDJ's corpus scale is hundreds, not millions, of papers, justifying Postgres full-text search over a dedicated search engine; authors submit PDFs; a single transactional email provider is available. Dependencies: AWS managed services (ECS, RDS, S3, CloudFront), Vercel, and a transactional email provider must remain available and within free/low-cost tier limits for the system to operate as specified.

3. Specific requirements

Each functional requirement is stated as "the system shall ...", is atomic (one testable behaviour per identifier), and carries the preconditions that must hold before it may fire, the postconditions guaranteed on success, and the acceptance criterion by which it is verified. Identifiers are preserved unchanged from the design specification; no requirement is renumbered. One requirement, **FR-25a**, is added beyond the original set and is marked **NEW** — it was surfaced by comparing the implemented lifecycle guard (`transitions.py` , which permits `WITHDRAWN` as a target state) against the implemented authorisation layer (`policies.py` , which has no action gating who may invoke that transition), a gap the original requirement (FR-25, which named the feature but not its authorisation mechanism) did not anticipate.

3.1 Functional requirements

Group A — Identity and access

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-01	The system shall let a person register an account and shall verify their email address before the account is usable.	Author, Reviewer	M	Email address not already registered	Account exists in an unverified state until the verification link is followed	Registration with a duplicate email is rejected; login is refused until verified
FR-02	The system shall authenticate a registered user and maintain a session that expires and can be explicitly revoked.	All authenticated	M	Account verified	A valid session token is issued; revoked or expired tokens are rejected on every subsequent call	A revoked refresh token fails all further use; an expired access token is rejected without a valid session
FR-03	The system shall let an Administrator assign and revoke roles, and a user may hold several roles concurrently.	Administrator	M	Target account exists	The account's role set reflects the change immediately	Assigning a second role does not remove the first; policy checks re-evaluate against the updated set on the next request

Group B — Submission intake and processing

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-04	The system shall let an Author create a submission comprising title, abstract, keywords, co-authors and a PDF file.	Author	M	Author is authenticated	Manuscript exists in DRAFT , then transitions to SUBMITTED	A manuscript with all required fields and a valid PDF reaches SUBMITTED ; the transition is illegal from any other state
FR-05	The system shall reject an uploaded file that fails type, size or integrity validation, and shall return a clear reason.	System	M	A file has been uploaded	Rejected files never reach persistent storage or a manuscript version	A renamed non-PDF file (validated by magic bytes, not extension) is rejected with a specific reason code
FR-06	The system shall generate an anonymised copy of every submitted document, distinct from the original.	System	M	A manuscript version has been uploaded and validated	Two document references exist per version: original and anonymised	The anonymised derivative has XMP and DocInfo metadata stripped; reviewers are never issued the original
FR-20	The system shall screen a submission for similarity against the internal corpus of previously submitted manuscripts.	System	S	Text has been extracted from the submission	A similarity report is persisted against the manuscript version	A near-duplicate submitted twice produces a report flagging the match; unrelated submissions produce no false positive on the seeded corpus

Group C — Screening and reviewer assignment

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-07	The system shall let an Editor or Editor-in-Chief screen a submission by sending it to review, requesting pre-review changes, or desk-rejecting it.	Editor	M	Manuscript is in UNDER_SCREENING	Manuscript moves to UNDER_REVIEW , REVISION_REQUESTED or DESK_REJECTED respectively	Any of the three legal targets is reachable only from UNDER_SCREENING ; no other actor role can invoke it
FR-08	The system shall recommend reviewers ranked by expertise match, excluding conflicts of interest and respecting reviewer capacity.	System	M	Manuscript is in UNDER_SCREENING or UNDER_REVIEW	A ranked candidate list is produced with a score and, for excluded reviewers, a stated reason	A reviewer sharing an author's affiliation, or already at capacity, never appears as a positive recommendation
FR-09	The system shall let an Editor invite reviewers, and may override any system recommendation.	Editor	M	Manuscript is in UNDER_SCREENING or UNDER_REVIEW	A review assignment is created in the INVITED state	An editor-chosen reviewer outside the recommended list can still be invited

Group D — Review and revision

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-10	The system shall let an invited reviewer accept or decline the invitation, optionally with a reason.	Reviewer	M	Assignment is in INVITED	Assignment moves to ACCEPTED or DECLINED	A declined invitation frees the reviewer's capacity and is visible to the editor with its reason

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-11	The system shall let a reviewer with an accepted assignment submit a structured review: criterion scores, an overall recommendation, comments to the author, and confidential comments to the editor.	Reviewer	M	Assignment is ACCEPTED ; manuscript is UNDER_REVIEW	A Review record is stored against the assignment; assignment moves to SUBMITTED	Confidential comments never appear in any author-facing view; a review cannot be submitted twice by the same reviewer for the same manuscript
FR-13	The system shall let the corresponding author submit a revised version together with a response-to-reviewers letter.	Author	M	Manuscript is in REVISION_REQUESTED ; actor is the corresponding author	Manuscript moves to RESUBMITTED , then to UNDER_REVIEW or UNDER_SCREENING at editorial discretion	A non-corresponding co-author's attempt to resubmit is denied; the letter is attached to the new version

Group E — Editorial decision and withdrawal

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-12	The system shall let an Editor or Editor-in-Chief record an editorial decision once the configured minimum number of reviews has been submitted.	Editor	M	Manuscript is REVIEWS_COMPLETED (or UNDER_SCREENING for desk rejection, which requires no reviews)	Manuscript moves to ACCEPTED , REJECTED or REVISION_REQUESTED	A decision attempted below quorum is refused with the current review count; desk rejection is legal only from UNDER_SCREENING
FR-25	The system shall let the corresponding author withdraw a submission before a decision is recorded.	Author	S	Manuscript is in SUBMITTED , UNDER_SCREENING , UNDER_REVIEW , REVIEWS_COMPLETED or REVISION_REQUESTED	Manuscript moves to WITHDRAWN , a terminal state	A withdrawal attempted from ACCEPTED or any other non-listed state is refused
FR-25a (NEW)	The system shall authorise withdrawal only for the corresponding author of the manuscript being withdrawn, via a dedicated ownership-checked action equivalent to how RESUBMIT is checked.	Author	M <i>(elevated — closes a live authorisation gap)</i>	Action.WITHDRAW exists in the policy layer with the same ownership predicate as Action.RESUBMIT	An actor who is not the corresponding author, including a co-author, cannot invoke the transition regardless of role	A non-corresponding-author holder of the AUTHOR role attempting withdrawal is denied

Group F — Publication and public archive

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-15	The system shall let the Editor-in-Chief compose an issue from accepted manuscripts and publish it.	Editor-in-Chief	M	Manuscripts are ACCEPTED ; actor holds EDITOR_IN_CHIEF	Manuscripts move ACCEPTED → SCHEDULED → PUBLISHED ; the issue becomes publicly visible	An Editor without EiC role cannot publish; a manuscript not in ACCEPTED cannot be added to an issue

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-16	The system shall let an unauthenticated reader browse issues and read published papers.	Reader	M	Issue is PUBLISHED	Paper content and metadata are served without requiring login	Anonymous request to a published paper's page succeeds; request to an unpublished manuscript is refused
FR-17	The system shall provide full-text search across published papers with filters.	Reader	M	At least one paper is published	Search returns ranked results within the performance bound of NFR-09	A query on a distinctive term present only in one seeded paper returns that paper first
FR-18	The system shall let a reader download a published paper's PDF.	Reader	M	Paper is published	The original (non-anonymised) PDF is served via the archive path	Download succeeds without authentication for any published paper
FR-21	The system shall let a reader export a paper's citation as BibTeX and RIS.	Reader	S	Paper is published	A correctly formatted citation file is returned in the requested format	The exported BibTeX parses under a standard BibTeX parser without error
FR-22	The system shall expose an OAI-PMH endpoint serving Dublin Core metadata for all published papers.	External system	S	At least one paper is published	Identify , ListMetadataFormats , ListIdentifiers , ListRecords and GetRecord all respond per OAI-PMH 2.0	A harvest run against the seeded archive completes with resumption tokens honoured correctly
FR-27	The system shall resolve a persistent identifier to its paper.	Reader	C	Identifier is DOI-shaped and assigned	Resolving the identifier returns (or redirects to) the paper	A well-formed but unassigned identifier returns a 404, not a server error

Group G — Transparency and communication

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-14	The system shall let an author view their own submissions and each one's status timeline.	Author	M	Actor is authenticated	Only manuscripts the actor authored are listed	An author cannot see another author's manuscript in this view, even by guessing its identifier
FR-19	The system shall let an Editor or Editor-in-Chief view the complete audit trail for a manuscript.	Editor, EiC	M	Actor holds EDITOR or EDITOR_IN_CHIEF	The full, sequence-ordered event chain for the manuscript is returned	A Reviewer or Author role attempting this view is denied; the chain's hash verification passes
FR-23	The system shall notify actors, in-app and by email where configured, of events affecting them.	System	S	An event affecting a specific actor has occurred	A notification record exists and, where email is configured, a message is queued for delivery	An editorial decision produces a notification visible to the affected author within the same request cycle

Group H — Administration, configuration and analytics

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-24	The system shall provide editorial analytics: throughput, decision mix, reviewer load and time-to-decision.	EiC, HoD	S	At least one full editorial cycle has completed	Aggregate figures are computed over the current dataset	Time-to-decision reported for a manuscript matches the difference between its submission and decision timestamps
FR-26	The system shall let the Editor-in-Chief configure journal policy: minimum reviews, review window and blinding model.	EiC	C	Actor holds EDITOR_IN_CHIEF	Subsequent decisions and assignments respect the updated configuration	Lowering the minimum-review threshold takes effect for manuscripts not yet at decision, not retroactively
FR-28	The system shall provide reviewer performance history and reliability indicators.	EiC	C	Reviewer has at least one completed or declined assignment	A per-reviewer summary (acceptance rate, on-time rate, average score deviation) is available	A reviewer who declines every invitation shows a 0% acceptance rate, not a null or error
FR-30	The system shall let an Administrator manage accounts: grant and revoke roles, set reviewer capacity, and deactivate accounts.	Administrator	M	Actor holds ADMINISTRATOR	The account's roles, capacity or activation state are updated and visible on the next read	The administrator role itself cannot be granted or revoked through this interface, and an administrator cannot deactivate their own account
FR-33	The system shall record a due date for each review assignment and mark an assignment overdue once that date passes without a submitted review.	Editor, EiC	S	A reviewer is assigned to a manuscript	The assignment carries a deadline and a server-computed overdue flag	A review submitted after its deadline is not flagged overdue; an assignment with no deadline is never flagged overdue

Group I — Publication economics and scholarly record

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-29	The system shall raise an article processing charge when a manuscript is accepted, and let the corresponding author settle it through a payment gateway.	Author, EiC	S	Manuscript is ACCEPTED, SCHEDULED or PUBLISHED	An invoice exists with a status of pending, paid or waived	No invoice exists before acceptance; a settled invoice cannot be waived, and a waived invoice cannot be verified
FR-31	The system shall issue a signed-style decision certificate for a manuscript whose final decision is recorded.	Editor, EiC	C	An accept or reject decision exists in the manuscript's event chain	A PDF stating the decision, the tracking code and the audit chain's head hash is returned	Requesting a certificate before any final decision is a 409, not an empty document
FR-32	The system shall report, at submission time, what the anonymiser removed from the uploaded document and what it could not remove.	Author	S	A document has been uploaded	The submission response carries a report of removed metadata keys and any author names still detectable in the body text	An author whose name appears in the manuscript body is told so at submission rather than after a reviewer sees it

ID	Statement	Actor	Pri	Preconditions	Postconditions	Acceptance criteria
FR-34	The system shall let a member of the public register an author account without editorial-office involvement.	Reader	S	The email address is not already registered	An account exists holding exactly the <code>AUTHOR</code> role, signed in	Registration cannot grant reviewer, editor or administrator roles under any request body

3.2 Non-functional requirements

Each non-functional requirement carries a stated **verification method** so that "non-functional" does not collapse into "unverifiable."

Security

ID	Requirement	Verification
NFR-01	Passwords are hashed with Argon2id; no credential is ever written to a log.	Code review; unit test asserting the hashing algorithm and asserting no credential-shaped value appears in captured log output
NFR-02	Every endpoint is authorised by an explicit policy; access is denied by default.	Integration test that walks the full route table and fails if any route lacks a policy check (design specification section 11)
NFR-03	Reviewer-facing payloads contain no author-identifying data.	Automated leak test enumerating every reviewer-facing endpoint, serialising against a fixture with sentinel author names, asserting no sentinel appears in output
NFR-04	Documents are served only via short-lived pre-signed URLs; storage buckets are private.	Integration test confirming direct bucket access is refused; manual verification of URL expiry
NFR-05	Authentication and submission endpoints are rate-limited.	Integration test exceeding the configured limit and asserting a 429 response
NFR-06	Dependencies are scanned; no known high-severity vulnerability is present at release.	<code>pip-audit</code> and <code>npm audit</code> as CI gates

Integrity

ID	Requirement	Verification
NFR-07	Audit events are append-only and hash-chained; any retrospective edit is detectable.	Property test over random legal event sequences confirming the chain always verifies; a tamper-detection test that mutates a stored event and confirms verification fails. Demonstrated in the current build by <code>backend/tests/unit/domain/test_hashchain.py</code> and the integration tests <code>test_append_only.py</code> and <code>test_chain_persistence.py</code> , including a PostgreSQL trigger rejecting <code>UPDATE / DELETE / TRUNCATE</code> on the event table (technical debt register TD-13)

Performance

ID	Requirement	Verification
NFR-08	Public archive pages respond within 500 ms at p95 under nominal load.	k6 smoke test against a deployed environment
NFR-09	Search returns within 800 ms at p95 over the seeded corpus.	k6 smoke test

Reliability

ID	Requirement	Verification
NFR-10	Upload processing is idempotent and retried with exponential backoff on failure.	Unit test on the retry policy; integration test replaying the same content checksum and asserting no duplicate side effect

Availability

ID	Requirement	Verification
NFR-11	The service exposes liveness and readiness probes; unhealthy tasks are replaced automatically.	Deployment verification against the ECS service's health-check configuration

Usability

ID	Requirement	Verification
NFR-12	The interface is responsive from 360 px width, keyboard-navigable, and meets WCAG 2.1 AA contrast.	Manual audit; automated <code>axe</code> accessibility check in CI

Maintainability

ID	Requirement	Verification
NFR-13	The domain layer imports no framework code.	Import-linter contract enforced as a CI gate
NFR-14	The domain and application layers hold at least 85% line coverage.	CI coverage gate. The delivered domain and application layers report 90.03% against the 85% gate, confirmed directly (<code>pytest -m "not integration" -cov=src/ugjcs/domain -cov=src/ugjcs/application</code> , run against this commit): 402 unit tests pass, and a separate run collects 84 integration tests against a real PostgreSQL container. Technical debt register TD-10 records the 85% figure as a floor, not a target, and TD-11 records that coverage alone is a weak signal — four mutations survived a 100%-covered suite until closed by targeted tests

Observability

ID	Requirement	Verification
NFR-15	Structured JSON logs carry correlation IDs; traces are exported.	Manual verification against a deployed environment

Portability

ID	Requirement	Verification
NFR-16	The entire infrastructure is reproducible from code.	<code>terraform apply</code> from a clean state, followed by <code>terraform destroy</code>

Compliance

ID	Requirement	Verification
NFR-17	Personal data is minimised; the audit log retains identity only where required.	Design review against the data model (design specification section 8)

4. System models

4.1 Manuscript lifecycle

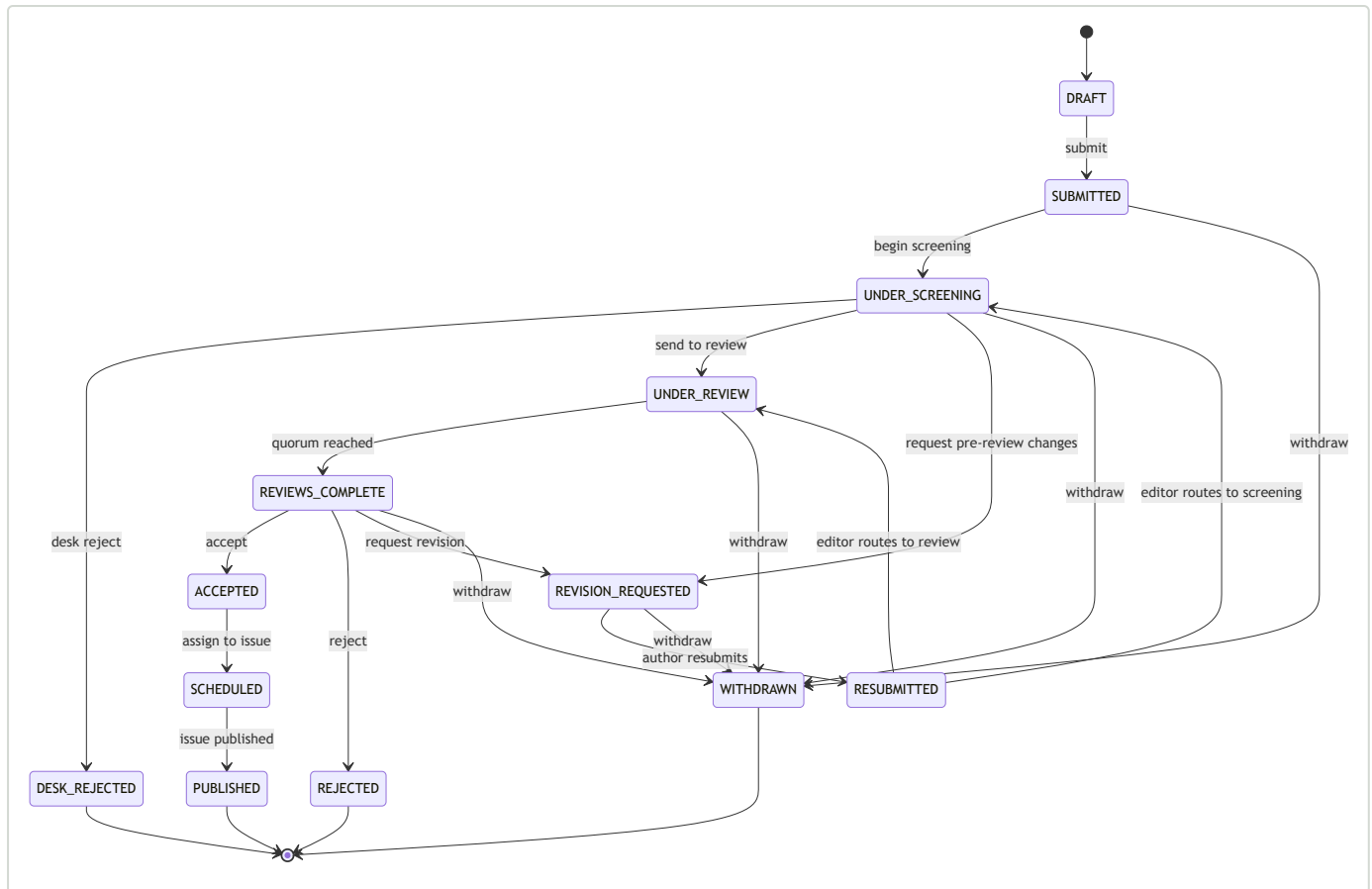
The lifecycle below is transcribed directly from `backend/src/ugjcs/domain/transitions.py` — the `LEGAL_TRANSITIONS` mapping is the executable, tested source of truth, not the narrative description in the design specification.

4.1.1 State table

Source state	Legal target states	Terminal?
DRAFT	SUBMITTED	No
SUBMITTED	UNDER_SCREENING , WITHDRAWN	No
UNDER_SCREENING	DESK_REJECTED , UNDER_REVIEW , REVISION_REQUESTED , WITHDRAWN	No
UNDER_REVIEW	REVIEWS_COMPLETE , WITHDRAWN	No
REVIEWS_COMPLETE	ACCEPTED , REJECTED , REVISION_REQUESTED , WITHDRAWN	No
REVISION_REQUESTED	RESUBMITTED , WITHDRAWN	No
RESUBMITTED	UNDER_REVIEW , UNDER_SCREENING	No
ACCEPTED	SCHEDULED	No
SCHEDULED	PUBLISHED	No
DESK_REJECTED	(none)	Yes
REJECTED	(none)	Yes
PUBLISHED	(none)	Yes
WITHDRAWN	(none)	Yes

No transition is legal out of a terminal state; this is asserted directly by `assert_legal`, which raises `IllegalTransitionError` for any pair not present in `LEGAL_TRANSITIONS`. `ACCEPTED` and `SCHEDULED` are deliberately excluded from the withdrawable set: once a manuscript is accepted, undoing that is editorial retraction — a different action, with its own notice obligations — and is not modelled as withdrawal.

4.1.2 State diagram



4.1.3 Spec-versus-code disagreement — RESUBMITTED routing

The design specification's section 6.2 diagram draws `RESUBMITTED` flowing directly into `REVIEWS_COMPLETE` ("`RESUBMITTED` $\leftarrow$... $\rightarrow$ `REVIEWS_COMPLETE`"), implying a resubmission closes the review round automatically. The implemented `LEGAL_TRANSITIONS` table instead routes `RESUBMITTED` to **either** `UNDER_REVIEW` **or** `UNDER_SCREENING`, at editorial discretion — a resubmission may need fresh review, or may need re-screening first, but never closes the review round by itself. **The code governs**, per this document's instruction: FR-13's postcondition and the state diagram above are written to the implemented behaviour, not the narrative diagram. This is a genuine, load-bearing disagreement between the two documents and is recorded here rather than silently resolved in one direction without comment.

4.2 Actor/use-case mapping

Actor	Use cases (effort estimation section 3)
Author	UC1, UC2, UC4, UC9 (as respondent is Reviewer; Author has no UC9 role), UC12, UC13, UC19, UC21 (recipient), UC23
Reviewer	UC1, UC2, UC9, UC10, UC21 (recipient)
Editor	UC2, UC3 (Administrator only, not Editor — see note), UC6, UC7, UC8, UC11, UC18, UC21 (recipient)
Editor-in-Chief	All Editor use cases, plus UC14, UC22, UC24
Administrator	UC3
Reader	UC15, UC16, UC17, UC19 (public export, distinct from FR-19's audit view), UC20 (indirectly, as the harvested party), UC25
CBAS college leadership	UC22 (read-only consumer)
Indexing service	UC20

Actor	Use cases (effort estimation section 3)
System (no human actor)	UC5, UC7 (computation), UC21 (dispatch)

Note: UC3 ("manage roles") is an Administrator-only use case per FR-03's actor column; it is listed once and not duplicated against Editor.

4.3 Authorisation matrix

Derived directly from `backend/src/ugjcs/domain/policies.py` — `_ROLE_GRANTS` for role-only actions, `_OWNERSHIP_ACTIONS` and `_can_view` for actions that also depend on the actor's relationship to a specific manuscript, and the policy module's own docstring for the one action (`REVIEW`) that is deliberately under-specified today.

Action	Author	Reviewer	Editor	Editor-in-Chief	Administrator
VIEW	Own manuscripts only	— (<i>no dedicated action; see gap below</i>)	Yes	Yes	Yes
SUBMIT	Yes	—	—	—	—
SCREEN	—	—	Yes	Yes	—
ASSIGN_REVIEWER	—	—	Yes	Yes	—
REVIEW	—	Yes (<i>role-only — no conflict-of-interest predicate; TD-02</i>)	—	—	—
DECIDE	—	—	Yes	Yes	—
RESUBMIT	Corresponding author only	—	—	—	—
PUBLISH	—	—	—	Yes	—
MANAGE_USERS	—	—	—	—	Yes
VIEW_AUDIT	—	—	Yes	Yes	— (<i>role grant exists; no route reaches it — section 5, FR-19</i>)
WITHDRAW (<i>FR-25a</i>)	Corresponding author only — implemented & tested	—	—	—	—

FR-25a is closed. `Action.WITHDRAW` now exists in `_OWNERSHIP_ACTIONS` alongside `RESUBMIT` (`policies.py`), and `POST /manuscripts/{tracking_code}/withdraw` (`manuscripts.py`) calls `authorize(actor, Action.WITHDRAW, manuscript)` before withdrawing.

`test_corresponding_author_may_withdraw_own_manuscript`, `test_another_author_may_not_withdraw_someone_elses_manuscript` and `test_a_listed_co_author_who_is_not_corresponding_may_not_withdraw` (`test_policies.py`), plus `test_the_corresponding_author_can_withdraw` and `test_a_co_author_who_is_not_corresponding_cannot_withdraw` (`test_manuscripts_router.py`), cover it end to end. The gap this row previously recorded — first noticed as a mismatch between `transitions.py` (which already permitted `WITHDRAWN`) and `policies.py` (which had no gate for it) — is resolved, not merely documented.

Known gaps still open in this matrix, cross-referenced to the technical debt register:

- `REVIEW` is granted to any actor holding the `REVIEWER` role with no check that they are not also an author of, or affiliated with, the manuscript in question (**TD-02** — critical, still open). The reviews API (`reviews.py`) adds one partial mitigation not present when this gap was first recorded: `_assigned_or_403` refuses a

reviewer who was never assigned to the manuscript. It does **not** stop a reviewer who *was* assigned, and who also happens to be a listed author, from reviewing their own work — `assign_reviewer` (`editorial.py`) never checks `author_ids` before creating an assignment. Because `Manuscript.record_review` also does not check the submitting reviewer's identity against an accepted assignment or against reviewers who already submitted (**TD-03** — critical, still open), a reviewer with an existing assignment could in principle call `POST /reviews/{tracking_code}/submit` twice and reach the quorum alone. Neither gap has been closed by the API layer; both remain exactly as critical as the technical debt register states.

- There is no Action connecting `blinding.blind()` to `policies.can()` — a caller must remember to call `blind()` before serving a reviewer, and nothing in the authorisation layer enforces it (**TD-07** — scheduled). In the delivered code, `reviews.py`'s `my_assignments` handler does call `blind()` correctly, and `test_blinding_leak.py` proves no author identity leaks through it today — but that is adapter discipline holding, which is precisely the weakness TD-07 names, not the gap being closed.
- The editorial event log has no blinded projection and is protected by `VIEW_AUDIT` policy alone rather than by the type system (**TD-06** — scheduled). No reviewer-facing path reads the log in the delivered system, so the risk TD-06 describes remains theoretical, exactly as originally recorded.

5. Requirements traceability matrix

Revised 2026-08-12 against the finished system (see the revision note on p.1). Status categories:

Implemented & tested (the requirement's behaviour is reachable end to end — domain rule, API route and, where the requirement names one, a frontend screen — and is covered by an automated test in the delivered codebase); **Partially implemented** (some layer of the requirement is built and tested but a named, specific piece is missing — never "mostly done" without saying what the remainder is); **Not implemented** (no code exists for this requirement's behaviour, stated as such rather than described euphemistically).

FR	Use case	Module / endpoint	Test	Status
FR-01	UC1	<code>application/identity.py</code> (<code>RegistrationService</code>) — account creation, email verification token issue and redemption	<code>test_identity.py</code> : <code>test_registering_creates_an_unverified_account_and_sends_one_message</code> , <code>test_a_valid_verification_token_verifies_the_account</code> , <code>test_a_verification_token_cannot_be_replayed</code> , <code>test_registering_an_existing_email_raises_and_sends_no_second_message</code>	Partially implemented. The service layer is fully built and tested, including duplicate-email and replay handling. But no <code>/auth/register</code> route exists (absent from the live <code>/openapi.json</code>) and the frontend has no registration screen — only <code>/login</code> . Accounts in the deployed system are provisioned by seed data (the judge accounts in <code>docs/06-testing-report.md</code> section 5), not self-registration. Missing: the API route and the frontend form that would call it.
FR-02	UC2	<code>api/routers/auth.py</code> (<code>/auth/login</code> , <code>/auth/refresh</code> , <code>/auth/logout</code> , <code>/auth/me</code>); <code>application/identity.py</code> (<code>SessionService</code>)	<code>test_auth_router.py</code> (unit, all four routes); integration <code>test_refresh_rotation.py</code> : <code>test_replaying_a_rotated_refresh_token_revokes_the_entire_family</code> , <code>test_roles_revoked_after_a_token_was_issued_take_effect_immediately</code> , <code>test_an_expired_refresh_token_is_refused</code>	Implemented & tested. Session issuance, explicit revocation (<code>/logout</code>), and expiry are all live and covered, including the security-sensitive case (stolen-refresh-token family revocation) against a real database.

FR	Use case	Module / endpoint	Test	Status
FR-03	UC3	domain/policies.py (Action.MANAGE_USERS , granted to Administrator); infrastructure/db/account_repository.py (role grant/revoke persistence)	test_policies.py (grant); integration test_account_repository.py : test_granting_a_role_and_saving_persists_it , test_revoking_a_role_and_saving_removes_it	Partially implemented. The authorisation grant and the persistence of a role change are both implemented and tested against a real database. No /admin API route or frontend screen exists to let an Administrator invoke this — confirmed absent from /openapi.json and from frontend/src/app/ , and stated explicitly in docs/06-testing-report.md section 5: "no UI surface exists for this yet."
FR-04	UC4	api/routers/manuscripts.py (POST /manuscripts , multipart with file); domain/manuscript.py ; domain/transitions.py (DRAFT→SUBMITTED); frontend app/author/submit/page.tsx	test_manuscripts_router.py : test_an_author_can_submit_a_manuscript_with_a_pdf ; test_transitions.py , test_manuscript.py	Implemented & tested , end to end: route, upload, lifecycle guard and the author's submission form all exist and are exercised by the cited tests plus the live UAT pass in section 5 of the testing report.
FR-05	UC4	application/documents.py (validate_document — magic-byte check, independent of client-supplied Content-Type)	test_documents.py : test_a_client_supplied_content_type_cannot_substitute_for_the_magic_number , test_content_exceeding_the_size_cap_is_rejected ; test_manuscripts_router.py : test_a_non_pdf_upload_is_rejected_with_415 , test_an_oversized_upload_is_rejected_with_413	Implemented & tested , at the unit level (validation logic) and the API level (415 / 413 responses).
FR-06	UC5	infrastructure/storage/anonymize.py (strip_pdf_metadata) ; wired into manuscripts.py 's _store_document , which writes both an original and an anonymised S3 key on every submit/resubmit	test_anonymize.py ; test_documents.py : test_original_and_anonymised_keys_differ_for_the_same_manuscript_and_version , test_keys_carry_no_title_or_author_identifying_text	Implemented & tested. Every submitted document produces a metadata-stripped derivative, distinct from the original, before either reaches storage. Note the scope of what "anonymised" means here — see section 7: XMP/DocInfo metadata is stripped; visible body text (a title or abstract that names the author) is not touched, which is FR-06 as designed, not a shortfall against it.

FR	Use case	Module / endpoint	Test	Status
FR-07	UC6	api/routers/editorial.py (POST /editorial/{tracking_code}/screen → begin_screening ; POST /editorial/{tracking_code}/decision with DESK_REJECT / SEND_TO_REVIEW / REQUEST_REVISION → the three FR-07 outcomes); domain/transitions.py ; domain/policies.py (Action.SCREEN , Action.DECIDE)	test_editorial_router.py : test_an_editor_can_begin_screening , test_a_decision_moves_the_manuscript_to_review , test_a_reviewer_cannot_record_a_decision ; test_transitions.py , test_policies.py	Implemented & tested. The three-way screening outcome is reached through the same /decision route FR-12 uses (one action, Action.DECIDE , gates the post-screening and post-review decisions alike) rather than a dedicated /screen -outcome route; /screen itself only performs the SUBMITTED-UNDER_SCREENING intake step. This is a routing detail, not a missing behaviour — all three outcomes are reachable and tested.
FR-08	UC7	api/routers/editorial.py (GET /editorial/{tracking_code}/reviewer-candidates , _match_score); domain/conflicts.py (exclusion_reason); api/schemas_analytics.py (RankedReviewerCandidateOut); frontend reviewer-candidate picker	test_editorial_router.py : test_reviewer_candidates_are_ranked_by_expertise_match_then_load , test_reviewer_candidates_lists_eligible_and_excluded_with_their_reasons , test_a_candidate_at_capacity_is_excluded , test_a_candidate_already_assigned_to_this_manuscript_is_excluded , test_reviewer_candidates_omits_unverified_inactive_and_non_reviewer_accounts , test_a_reviewer_cannot_see_reviewer_candidates	Implemented & tested. A ranked candidate list exists: match_score counts the manuscript's keywords appearing in a reviewer's expertise , candidates sort by match then by load, and exclusion_reason (a pure function) marks shared affiliation, existing assignment on this manuscript, and capacity exhaustion. Excluded candidates are returned with their reason rather than filtered away, so an editor can see why an obvious name is unavailable. Matching is keyword overlap, not TF-IDF or an assignment-optimisation algorithm; that is a simpler mechanism than the design specification sketched, and it is a deliberate one, since the corpus is far too small for term weighting to mean anything.
FR-09	UC8	api/routers/editorial.py (POST /editorial/{tracking_code}/reviewers); domain/policies.py (Action.ASSIGN_REVIEWER)	test_editorial_router.py : test_assigning_a_reviewer_records_it , test_assigning_a_reviewer_to_a_missing_manuscript_is_404 ; integration test_review_assignment_repository.py : test_an_assignment_is_visible_to_both_parties , test_assigning_the_same_reviewer_twice_is_rejected	Implemented & tested for the invite mechanism itself. The "may override any system recommendation" clause is now genuinely exercised rather than vacuous: FR-08's ranked candidate list exists, marks excluded reviewers with a reason instead of hiding them, and the editor remains free to assign any reviewer_id regardless of what the ranking said. assign_reviewer still does not check the invited reviewer against author_ids (TD-02) before creating the assignment, so the exclusion is advisory at the candidate-list layer rather than enforced at the assignment layer.

FR	Use case	Module / endpoint	Test	Status
FR-10	UC9	(none)	(none)	Not implemented. AssignmentStatus declares INVITED / ACCEPTED / DECLINED / SUBMITTED / EXPIRED, but the assignment repository writes the literal string "assigned" on creation (assignment_repository.py) — a value that is not even one of the enum's own members — and no route lets a reviewer accept or decline. A reviewer's assignment is usable immediately; there is no invitation-response step at all.
FR-11	UC10	api/routers/reviews.py (POST /reviews/{tracking_code}/submit); api/schemas.py (SubmitReviewRequest: four 1-5 criterion scores, comments_to_author, confidential_comments_to_editor; ReviewOut); api/routers/editorial.py (GET /editorial/{tracking_code}/reviews, the only route returning confidential comments); domain/manuscript.py (record_review, quorum counter)	test_reviews_router.py: test_submitting_a_review_counts_it_and_records_the_content, test_a_score_outside_one_to_five_is_rejected, test_submitting_without_an_assignment_is_forbidden; test_editorial_router.py: test_an_editor_can_list_the_reviews_recorded_for_a_manuscript, test_an_editor_sees_the_confidential_comments_a_reviewer_wrote; integration test_review_assignment_repository.py: test_marking_submitted_records_the_review_content	Implemented & tested. SubmitReviewRequest carries four criterion scores (originality, rigour, clarity, significance), each bounded 1 to 5 by Pydantic so an out-of-range score is a 422 before any handler runs, and the comment field is split into comments_to_author and confidential_comments_to_editor. The confidential channel is now enforced structurally rather than vacuously: ReviewOut is built by exactly one route, gated on Action.DECIDE, which no author-reachable or reviewer-reachable route carries. The quorum-counting defect (TD-03) is still unresolved; see section 4.3.
FR-12	UC11	api/routers/editorial.py (POST /editorial/{tracking_code}/decision); domain/transitions.py; domain/policies.py (Action.DECIDE)	test_editorial_router.py: test_a_decision_moves_the_manuscript_to_review, test_a_reviewer_cannot_record_a_decision; test_transitions.py, test_manuscript.py: test_a_decision_requires_the_minimum_review_count	Implemented & tested. Quorum enforcement and role gating both hold; quorum correctness still depends on the unresolved TD-03 defect in FR-11, which is a defect in what feeds the count, not in this requirement's own enforcement of it.

FR	Use case	Module / endpoint	Test	Status
FR-13	UC12	api/routers/manuscripts.py (POST /manuscripts/{tracking_code}/resubmit , multipart with response_to_reviewers); domain/policies.py (Action.RESUBMIT , ownership-checked)	test_manuscripts_router.py : test_the_corresponding_author_can_resubmit_with_a_revised_file , test_a_stranger_cannot_resubmit_someone_elses_manuscript , test_resubmitting_a_non_pdf_is_rejected ; test_manuscript.py : test_resubmission_increments_the_version_and_resets_review_count	Implemented & tested , including the response-to-reviewers letter and the ownership check.
FR-14	UC13	api/routers/manuscripts.py (GET /manuscripts/mine); domain/policies.py (Action.VIEW , _can_view); frontend app/author/page.tsx , app/author/[trackingCode]/page.tsx	test_manuscripts_router.py : test_mine_lists_only_the_callers_manuscripts , test_retrieving_someone_elses_manuscript_is_forbidden ; live UAT (docs/06-testing-report.md section 5): ten submissions listed with correct tracking codes and statuses	Partially implemented. An author's own-manuscripts list, correctly scoped and rendered, is implemented, tested and confirmed live. The status timeline clause is not met literally: ManuscriptOut carries only the manuscript's <i>current</i> status , not a history of transitions — the same underlying gap as FR-19 (no event/audit data is exposed on the wire at all, by the deliberate design recorded in schemas.py 's ManuscriptOut docstring). An author sees where a manuscript is now, not the sequence of states it passed through.
FR-15	UC14	api/routers/editorial.py (POST /editorial/{tracking_code}/schedule , POST /editorial/{tracking_code}/publish); domain/transitions.py (ACCEPTED→SCHEDULED→PUBLISHED); domain/policies.py (Action.PUBLISH , EiC-only)	test_editorial_router.py : test_the_editor_in_chief_can_schedule_an_accepted_manuscript , test_the_editor_in_chief_can_publish_a_scheduled_manuscript , test_a_plain_editor_cannot_schedule , test_a_plain_editor_cannot_publish , test_publishing_without_scheduling_first_is_a_conflict , test_scheduling_the_same_volume_and_number_twice_yields_the_same_issue_id	Implemented & tested. Issue is not a persisted aggregate — an IssueId is derived deterministically from (volume , number) (domain/ids.py , mint_issue_id) rather than looked up — a documented simplification, not a gap against this requirement's testable behaviour. docs/06-testing-report.md section 4.5 records that these two routes, plus resubmission, were at one point fully domain-tested but reachable by no route at all; that gap is closed and is exactly what the cited tests now pin.

FR	Use case	Module / endpoint	Test	Status
FR-16	UC15	api/routers/archive.py (GET /archive , GET /archive/{tracking_code} — no auth dependency); frontend app/(public)/papers/[trackingCode]/page.tsx	test_archive_router.py : test_the_archive_requires_no_authentication , test_retrieving_a_published_paper_by_tracking_code , test_an_unpublished_manuscript_is_not_found_via_the_archive ; route audit test_the_archive_prefix_genuinely_has_no_authorization_dependency	Implemented & tested , including the negative case (an unpublished manuscript is not reachable through this path) and the route-audit proof that the entire /archive prefix is genuinely, not accidentally, public.
FR-17	UC16	infrastructure/storage/fulltext.py (PDF text extraction); infrastructure/db/models.py (stored tsvector column); infrastructure/db/repository.py (search_published , websearch_to_tsquery ranked by ts_rank , ts_headline snippets); api/routers/archive.py (GET /archive/search); api/schemas_wave2.py (ArchiveSearchResult Out); frontend app/(public)/search/page.tsx	test_archive_search_fulltext.py : test_a_fulltext_match_carries_a_snippet_of_content , test_a_title_match_carries_a_null_snippet , test_a_keyword_match_is_found_with_a_null_snippet , test_no_match_anywhere_returns_an_empty_list , test_publishing_extracts_the_pdf_text_into_the_search_column , test_a_missing_or_unreadable_document_never_blocks_publishing ; integration test_fulltext_search.py , test_archive_queries.py	Implemented & tested. Search is PostgreSQL full-text over a stored tsvector covering title, abstract, keywords and the extracted body text of the published PDF, ranked by ts_rank , with ts_headline snippets returned when the match landed in the body. Body text is extracted at publication and a failed extraction never blocks the publish. Still not met: the "with filters" clause, since no keyword or date filter exists. NFR-09's 800 ms p95 bound remains unverified, because no load or performance test exists in this project (docs/06-testing-report.md section 6).
FR-18	UC17	api/routers/archive.py (GET /archive/{tracking_code}/document , no auth dependency); api/schemas.py (ArchivePaperOut.has_document , DocumentUrlOut); frontend app/(public)/papers/[trackingCode]/page.tsx (inline PDF viewer)	test_archive_router.py : test_a_published_papers_document_is_downloadable_without_authentication , test_archive_entries_report_whether_a_document_is_attached , test_downloading_a_missing_papers_document_is_404 , test_downloading_when_no_document_was_ever_attached_is_404 ; route audit test_the_archive_prefix_genuinely_has_no_authorization_dependency	Implemented & tested. A public, unauthenticated route serves a pre-signed URL for the original document of a published manuscript only. has_document on the archive shape lets a reader's page decide whether to render a viewer without first attempting a download. The route sits inside the /archive prefix, which the route audit proves carries no authorisation dependency anywhere.

FR	Use case	Module / endpoint	Test	Status
FR-19	UC18	domain/hashchain.py ; PostgreSQL append-only triggers (Alembic migration); api/routers/archive.py (GET /archive/{tracking_code}/provenance); api/schemas_scholarly.py (ProvenanceOut, ProvenanceEventOut); api/routers/certificate.py (decision certificate carrying the chain head)	test_hashchain.py; test_provenance_router.py: test_an_intact_chain_verifies_and_reports_its_head_hash, test_a_tampered_interior_event_is_reported_as_not_intact, test_event_payloads_and_actor_ids_are_never_exposed, test_provenance_for_an_unpublished_manuscript_is_404; integration test_append_only.py: test_updating_an_event_is_rejected_by_the_database, test_truncating_the_event_log_is_rejected; test_chain_persistence.py	Implemented & tested. The chain was already enforced at the database boundary; GET /archive/{code}/provenance now surfaces it, and goes further than the requirement asked by making verification public rather than editor-only: anyone can recompute a published paper's chain from genesis and see intact, the head hash, and each event's type, timestamp and 8-character hash prefix. Payloads and actor_id are deliberately withheld, because a REVIEW_SUBMITTED payload names its reviewer and a public endpoint must not hand out even a pseudonymous handle. What intact does not prove (tail truncation, a forged event appended through the normal path, a history rebuilt wholesale from genesis) is stated in TD-04 and in the endpoint's own contract.
FR-20	UC5	(none)	(none)	Not implemented. No similarity-screening, MinHash, or LSH code exists anywhere in the delivered backend.
FR-21	UC19	application/scholarly.py (bibtex_citation, ris_citation, fake_doi, _citation_key); api/routers/archive.py (GET /archive/{tracking_code}/citation?format=); frontend components/paper-scholarly.tsx (CitationRow)	test_citation_router.py: test_a_bibtex_citation_is_well_formed_plain_text, test_a_ris_citation_is_a_journal_record_with_one_auth_tag_per_author, test_archive_entries_carry_the_doi_shaped_identifier, test_an_unknown_citation_format_is_422, test_a_missing_citation_format_is_422, test_a_citation_for_an_unpublished_manuscript_is_404	Implemented & tested. Both formats are served as text/plain from one route, with an unknown or missing format answered 422 rather than silently defaulting. Each paper also carries a DOI-shaped identifier derived from its tracking code. The identifier is not registered : 10.55555 is a documented fake registrant prefix (section 4.2) and resolving it at doi.org fails by design. Neither citation carries a publication date, because the domain stores none; that omission is the same underlying gap as FR-14's missing timeline.
FR-22	UC20	(none)	(none)	Not implemented. No OAI-PMH endpoint exists.
FR-23	UC21	infrastructure/email/logging_sender.py (LoggingEmailSender)	(none beyond FR-01's registration tests, which exercise the logging stub incidentally)	Not implemented , beyond a stub. LoggingEmailSender only logs a registration verification link — it does not send email, and there is no in-app notification record of any kind, no delivery for editorial events (decisions, assignments, publication), and no notification data model at all. Named explicitly in the design spec as future evolution; it remains that.

FR	Use case	Module / endpoint	Test	Status
FR-24	UC22	api/routers/editorial.py (GET /editorial/analytic s , gated on Action.VIEW_AUDIT); api/schemas_analyt ics.py (EditorialAnalytics0 ut , PipelineCounts , MonthlySubmissions); frontend app/editor/analytic s/page.tsx	test_editorial_analytics .py : test_analytics_reports_p ipeline_months_rate_and_ averages , test_analytics_over_an_e mpty_desk_reports_zeros_ and_nulls_not_zero_rates , test_a_reviewer_may_not_ read_analytics ; integration test_editorial_queries.p y	Implemented & tested. Throughput (pipeline counts by current status), decision mix (acceptance rate), and time-to-decision and review-turnaround averages are computed from the editorial event chain rather than from mutable manuscript rows, so the numbers derive from the same append-only record the audit guarantee covers. One rule governs every aggregate: an average or rate over an empty denominator is null , never 0 , so "no decisions yet" stays distinguishable from "decisions arrive instantly". The pipeline view drops draft (never reached the desk) and folds desk_rejected into rejected .
FR-25	UC23	api/routers/manuscr ipts.py (POST /manuscripts/{tracki ng_code}/withdraw); domain/transitions. py (WITHDRAWN reachable from five states); domain/policies.py (Action.WITHDRAW , ownership-checked — see FR-25a)	test_manuscripts_router. py : test_the_corresponding_a uthor_can_withdraw , test_a_co_author_who_is_ not_corresponding_cannot_ withdraw , test_withdrawing_a_missi ng_manuscript_is_404 ; test_transitions.py	Implemented & tested , now including the authorisation half this line previously lacked — see FR-25a.
FR-25a	UC23	domain/policies.py (Action.WITHDRAW in _OWNERSHIP_ACTIONS); api/routers/manuscr ipts.py (authorize(actor , Action.WITHDRAW , manuscript))	test_policies.py : test_corresponding_autho r_may_withdraw_own_manus cript , test_another_author_may_ not_withdraw_someone_els es_manuscript , test_a_listed_co_author_ who_is_not_corresponding _may_not_withdraw	Implemented & tested — this gap is closed. The authorisation predicate this line was added to demand now exists, is wired into the live route, and is directly tested for both the positive case and two distinct negative cases (a stranger; a non-corresponding co-author). Section 4.3 records this as resolved rather than merely documented.
FR-26	UC24	(none)	(none)	Not implemented. Deferred (Could-have; effort estimation section 8.1) — no policy-configuration route or UI exists.

FR	Use case	Module / endpoint	Test	Status
FR-29	(no dedicated UC — added after the original use-case set)	api/routers/billing.py (all four routes); api/schemas_wave2.py (ApcInvoiceOut, BillingInitializeOut, BillingVerifyOut); application/ports.py (PaymentGateway, ApcInvoiceRecord); infrastructure/config.py (Paystack key, optional); frontend/components/apc-panel.tsx	test_billing_router.py: test_an_accept_decision_opens_a_pending_invoice_at_the_default_tariff, test_a_non_accept_decision_opens_no_invoice, test_a_repeated_accept_path_never_double_bills, test_the_corresponding_author_can_read_their_invoice, test_a_co_author_cannot_read_the_invoice, test_mock_mode_initialize_settles_the_invoice_and_says_so, test_real_mode_initialize_returns_the_checkout_url_and_stores_the_reference, test_only_the_corresponding_author_may_initialize, test_initializing_a_settled_invoice_is_a_conflict; integration test_invoice_repository.py	Implemented & tested. An invoice opens automatically on an accept decision and only on an accept decision, at a default tariff, and the accept path is idempotent so a repeated transition never double-bills. Amounts are integer pesewas throughout, never floats. Mock mode is the default: with no Paystack secret key configured, initialize settles on the spot and answers mock: true, so the flow is demonstrable without a card; with a key configured it returns a real checkout URL and the charge is confirmed only by a later verify. Waiving is Editor-in-Chief only and is checked directly rather than by borrowing Action.PUBLISH, since waiving a charge and publishing a paper are different authorities that happen to sit with the same person.
FR-30	(no dedicated UC — added after the original use-case set)	api/routers/admin.py (all four routes, gated on Action.MANAGE_USERS); api/schemas_wave2.py (AdminAccountOut, RoleChangeRequest, CapacityChangeRequest, ActiveChangeRequest); frontend/app/admin/page.tsx	test_admin_router.py: test_every_admin_route_is_forbidden_to_non_administrators, test_admin_routes_require_authentication, test_the_roster_lists_every_account_with_its_administrative_shape, test_granting_the_reviewer_role_adds_it, test_revoking_a_held_role_removes_it, test_the_administrator_role_can_be_neither_granted_nor_revoked, test_capacity_can_be_set_within_one_to_ten, test_capacity_outside_one_to_ten_is_a_422, test_an_administrator_cannot_deactivate_and_reactivate_another_account	Implemented & tested. Action.MANAGE_USERS was defined in the policy layer from the start with no route exercising it; this closes that. Two refusals are enforced in the router rather than the domain because both protect the console from itself: the administrator role can be neither granted nor revoked through the API (403), and an administrator cannot deactivate their own account (409). Together they stop the last administrator locking everyone out.

FR	Use case	Module / endpoint	Test	Status
FR-31	(no dedicated UC — added after the original use-case set)	api/routers/certificate.py (GET /editorial-certificate/{tracking_code}, gated on Action.DECIDE); infrastructure/storage/certificate_pdf.py	test_certificate_router.py: test_an_editor_receives_a_pdf_certificate, test_the_editor_in_chief_may_also_fetch_a_certificate, test_the_certificate_never_names_a_reviewer_or_leaks_confidential_comments; test_certificate_auth.py	Implemented & tested. A generated PDF stating the final decision, the tracking code and the audit chain's head hash, so a decision can be attested outside the portal. Requesting one before any <code>accept</code> or <code>reject</code> decision is a 409 rather than an empty document. The certificate names no reviewer and carries no confidential comment, which is asserted directly rather than left to inspection.
FR-32	(no dedicated UC — added after the original use-case set)	infrastructure/storage/preflight.py (PdfInspection); api/schemas/scholarly.py (AnonymisationReport, ManuscriptSubmissionOut); api/routers/manuscripts.py (_submission_response); frontend submission form	test_submission_preflight.py: test_the_submission_response_reports_the_stripped_docinfo_keys, test_an_author_name_in_the_body_text_is_flagged, test_a_resubmission_also_carries_the_preflight_report, test_the_response_still_carries_every_manuscript_out_field	Implemented & tested. This is the visible half of TD-05. Metadata stripping cannot remove a name printed in the body text, so rather than leave that silent, submission and resubmission now return what was removed (DocInfo keys, XMP) and which author names were still found in the extracted text. <code>author_names_in_body</code> is an honest partial detector: an empty list means nothing was found, never that the document is proven clean. The response is a subclass of <code>ManuscriptOut</code> , so every existing consumer keeps every field it relied on.
FR-33	(no dedicated UC — added after the original use-case set)	api/routers/editorial.py (GET /editorial/{tracking_code}/assignments); api/schemas/analytics.py (AssignmentDeadlineOut); Alembic <code>due_at</code> migration; frontend editor assignments panel	test_editorial_analytics.py: test_assignments_lists_deadlines_names_and_the_overdue_flag, test_an_assignment_before_its_deadline_is_not_overdue, test_assignments_for_a_missing_manuscript_is_404; integration test_due_at_migration.py	Implemented & tested. <code>overdue</code> is computed server-side so every consumer agrees on one rule: not yet submitted, and <code>due_at</code> in the past. A review submitted late is never flagged overdue, and a null deadline never is either. The model carries <code>reviewer_name</code> , which is correct rather than a leak: the blind is author to reviewer, not editor to reviewer, and the route is gated on <code>Action.ASSIGN_REVIEWER</code> , which no author-reachable route carries.
FR-34	(no dedicated UC — added after the original use-case set)	api/routers/auth.py (POST /auth/register, RegisterRequest); application/registration.py (RegistrationService); frontend <code>app/login/page.tsx</code> (sign-up tab)	test_auth_router.py: test_registering_creates_a_verified_author_and_sings_in, test_registering_never_grants_an_editorial_role	Implemented & tested. The route grants exactly <code>Role.AUTHOR</code> and ignores any role the request body might carry, which is asserted directly. Email delivery is mocked (<code>LoggingEmailSender</code> logs the verification link rather than sending it), so the account is verified immediately and signed in; that shortcut is honest about itself and is the same stub FR-23 records as unbuilt. Password policy is length-based, enforced in <code>RegistrationService</code> .
FR-27	UC25	(none)	(none)	Not implemented. Deferred (Could-have) — no persistent-identifier resolution route exists.

FR	Use case	Module / endpoint	Test	Status
FR-28	(no dedicated UC — reporting over UC10/UC22 data, effort estimation section 3)	api/routers/editorial.py (GET /editorial/reviewer-performance , gated on Action.VIEW_AUDIT); api/schemas_analytics.py (ReviewerPerformance Out)	test_editorial_analytics.py : test_reviewer_performance_reports_workload_and_turnaround_per_reviewer , test_a_reviewer_may_not_read_reviewer_performance	Implemented & tested. Per reviewer: active assignments against capacity, reviews completed, average turnaround in days, and last activity. Turnaround and last activity are <code>null</code> until a first review completes, for the same reason the analytics aggregates are: an editor has to tell "new to the pool" apart from "turns reviews around instantly". The requirement's "acceptance rate" clause is not met, and cannot be, because FR-10's invitation lifecycle does not exist, so no reviewer has ever declined anything to compute a rate from.

Reading this matrix honestly. Of the 35 requirement lines (FR-01 to FR-34 plus FR-25a), **24 are implemented and tested end to end** (FR-02, FR-04, FR-05, FR-06, FR-07, FR-08, FR-09, FR-11, FR-12, FR-13, FR-15, FR-16, FR-18, FR-19, FR-21, FR-24, FR-25, FR-25a, FR-29 to FR-34), **5 are partially implemented with a specifically named remainder** (FR-01, FR-03, FR-14, FR-17, FR-28), and **6 are genuinely not implemented** (FR-10, FR-20, FR-22, FR-23, FR-26, FR-27).

FR-10, the reviewer invitation lifecycle, is the absence that propagates furthest: because no reviewer can decline, FR-28 cannot report an acceptance rate, and an assignment takes effect the moment it is created. The other five unbuilt requirements stand as the technical debt register and the design specification's future-evolution section describe them.

6. Requirements prioritisation

Prioritisation uses MoSCoW, carried unchanged from the design specification (section 5) and made authoritative by the effort estimation document's scope decision (section 8):

Priority	Use cases	Decision
Must	UC1–UC18	Implemented to production quality
Should	UC19–UC23	Implemented only if Must-have work completes early
Could	UC24, UC25	Deferred; recorded as technical debt with a repayment plan

How the estimate drove the cut. Use Case Points sized the Must-have subset alone at 188 UUCP → 125.7 UCP → 2,514 person-hours (effort estimation section 6), against a 48-hour development window — 1.9% of the Must-have estimate. COCOMO II Early Design, an independent cross-check on different inputs (source lines of code and process/product ratings rather than actor and transaction counts), priced the full system at approximately 7,170 person-hours. The two methods disagree by roughly 2.2× on the exact figure but agree on the order of magnitude: both place the full system in the one-to-four-person-year range, nearly two orders of magnitude beyond the available window (effort estimation section 7). That agreement, not either figure's precision, is what forced a MoSCoW cut rather than an attempt to build everything thinly. Should-have items are attempted only after every Must-have item reaches production quality; Could-have items (FR-26, FR-27, FR-28) are not attempted within the window under any circumstance.

Consequence for what exists today. At the time this document was first written, the domain layer was complete while the application, API and frontend layers were still planned — consistent with, not contrary to, the MoSCoW cut: the cut governs *what* is built, not the *order* in which layers within a use case are built, and a hexagonal architecture's domain core is the natural first layer to complete and verify in isolation (design specification section 7.1). That sequencing has since played out as intended: section 5's revised traceability matrix shows the API and frontend layers have caught up for most of the Must-have set (12 of 29 requirement

lines implemented and tested end to end, domain through frontend). Where a gap remains within the Must-have set — FR-08's reviewer matching and FR-19's audit-trail route being the clearest examples — it is a gap in a specific layer or route, not evidence that an entire layer was skipped; section 5 states each one by name rather than folding them back into a single "still catching up" claim.

7. Constraints and limitations

This system does not guarantee everything its feature list might suggest. The following limitations are stated plainly, each cross-referenced to `docs/04-technical-debt-register.md`.

- Double-blind review is not text-scrubbed — implemented as designed, with a limit that is now demonstrated in production, not merely anticipated.** Anonymisation strips XMP and DocInfo metadata from every submitted document (FR-06, live at `POST /manuscripts` and `POST /manuscripts/{tracking_code}/resubmit`, `infrastructure/storage/anonymize.py::strip_pdf_metadata`) and omits author fields from the `BlindedManuscript` type by construction (`blinding.py`), but `title`, `abstract` and `keywords` are copied to reviewers **verbatim**. An author who writes their own name into the title, or an abstract that says "extending our earlier work in [Obeng 2025]", reaches the reviewer unchanged. Double-blind integrity therefore depends partly on author compliance, not entirely on the system (**TD-05**, scheduled). This is unchanged by the API and frontend build — it was always the intended scope of FR-06, not a shortfall the implementation introduced.
- The audit trail has no external anchor.** Hash chaining (`NFR-07`, `hashchain.py`) detects alteration, reordering and removal *within* the chain, and a PostgreSQL trigger blocks `UPDATE`, `DELETE` and `TRUNCATE` on the event table — both properties are proven against a real database by `test_append_only.py` and `test_chain_persistence.py`. But truncation of the tail, a forged event appended through the legitimate API, or a wholly fabricated history rebuilt from the genesis hash, are **undetectable by the application alone** — there is no periodically published, independently held checkpoint to compare against (**TD-04**, scheduled). Deployment did not change this: no checkpoint-publishing mechanism was added, and — see FR-19 above — there is currently no API route at all through which anyone could view the chain to notice tampering even if the chain itself remained sound.
- A reviewer's conflict of interest is not checked by the authorisation layer — and this is now a live gap, not a theoretical one.** `Action.REVIEW` is granted to any actor holding the `REVIEWER` role, with no per-manuscript predicate excluding authors or affiliated actors (**TD-02**, critical). `POST /editorial/{tracking_code}/reviewers` creates an assignment without checking the reviewer against the manuscript's `author_ids`, and `POST /reviews/{tracking_code}/submit` counts a submission without checking it against reviewers who already submitted (**TD-03**, critical): a reviewer who also holds the `AUTHOR` role, or one who is assigned and calls `submit` twice, could in principle close a review round alone. Both routes are built and deployed today — the gap this bullet describes is no longer a forecast about what a future endpoint might do.
- Withdrawal authorisation now exists and is enforced — this bullet previously recorded a gap that is closed.** The lifecycle permits `WITHDRAWN` from five states (`transitions.py`), and `Action.WITHDRAW` (`policies.py`) now gates `POST /manuscripts/{tracking_code}/withdraw` on the actor being the manuscript's corresponding author, exactly as FR-25a specified. See section 4.3 and section 5 (FR-25a) for the tests that prove it. This entry is retained here, rather than deleted, so the document's history — a gap identified, then closed — stays legible.
- The editorial event log has no blinded view.** `EditorialEvent` carries `actor_id` and an editor's free-text rationale; there is no `BlindedEvent` projection, so a reviewer-facing audit view, if ever built, must not simply reuse `VIEW_AUDIT`'s current shape without one (**TD-06**, scheduled).
- The current state is materialised, not derived by event replay.** The hybrid design (event log plus a materialised `status` column) is a deliberate trade-off against full event sourcing, mitigated by routing every state change through one `_transition` method. Two representations of the same fact could in principle diverge if that discipline is ever broken (**TD-09**, acceptable).

- **Coverage is a floor, not a proof of correctness.** The CI gate at 85% (NFR-14) is a regression floor, not a target (**TD-10**); `make check` reports **90.03%** on the current commit against that 85% gate — 402 unit tests pass, plus 84 integration tests against a real PostgreSQL container, run separately (confirmed directly against this commit; see NFR-14). Separately, mutation testing during final review found defects surviving a suite at 100% line coverage — including deletion of the single line that makes the hash chain a chain — which coverage alone did not catch (**TD-11**, acceptable, provided compensated by targeted review; systematic mutation testing is recorded as future evolution, not yet in CI).
- **No reviewer accept/decline step exists (FR-10).** An assignment is usable by the reviewer immediately on creation; the `AssignmentStatus` vocabulary declares `INVITED / ACCEPTED / DECLINED`, but nothing in the delivered code writes or reads those states, and no route lets a reviewer decline. See section 5, FR-10.
- **A published paper cannot be downloaded by an anonymous reader (FR-18).** Document storage, upload, and authenticated download all exist (FR-04/FR-06), but no public, unauthenticated route serves a published manuscript's PDF, and the public paper page has no download link. This is the one gap this revision found that was not previously recorded anywhere in this project's documentation — see section 5, FR-18.
- **AWS access currently uses root credentials.** The toolchain authenticates as the AWS account root user rather than a least-privilege IAM principal (**TD-01**, critical). Deployment to App Runner, S3 and the rest of the live infrastructure has already happened under this arrangement, mitigated only by keeping the root keys off CI and used solely from the developer's own workstation (TD-01's full entry). That mitigation reduces exposure; it does not resolve the underlying debt, which remains the highest-priority item in the technical debt register.
- **Explicit out-of-scope items** (design specification section 4.2), restated here as hard limits rather than soft gaps: no copy-editing or typesetting workflow; no production-quality PDF galley generation; no multi-journal tenancy; no ORCID federation; identifiers are DOI-*shaped* but not registered with Crossref; similarity screening (FR-20, not implemented, see section 5) would in any case be against the internal corpus only, never the open web; email deliverability is not implemented at all today (FR-23), let alone guaranteed through a transactional provider; OAI-PMH (FR-22) is likewise not implemented. This list is not exhaustive of section 5's not-implemented rows, only a restatement of the ones the original specification called out as deliberately out of scope.
- **One item has left this list.** Article processing charges were originally declared out of scope and are now built (FR-29). The scope line that has *not* moved is the one that matters for anyone reading a charge on screen: the default configuration uses a mock gateway that settles an invoice without contacting Paystack, and a real Paystack integration is exercised only when a secret key is configured. No money has moved through this system, and the mock/real distinction is reported to the caller on every initialisation rather than inferred.
- **Requirements traceability reflects the finished system, not the pre-implementation build.** section 5 of this document, re-reconciled against the running code on 2026-08-14, shows 24 of 35 requirement lines implemented and tested end to end, 5 partially implemented with a named remainder, and 6 genuinely not implemented. The matrix has now been revised twice against the code rather than against intent, and both revisions moved lines in both directions: the second one had to correct rows that understated the system as much as the first corrected rows that overstated it. Section 5's closing paragraph is explicit that none of this is a claim of completeness.